






Apple’s photographs, in matching rounded frames. The pair and the closed case are from apple.com/airpods-5. The close-up is the speaker mesh from the Apple Store gallery. On the plates that follow, a thin line runs from the label to the part.

# AirPods 5

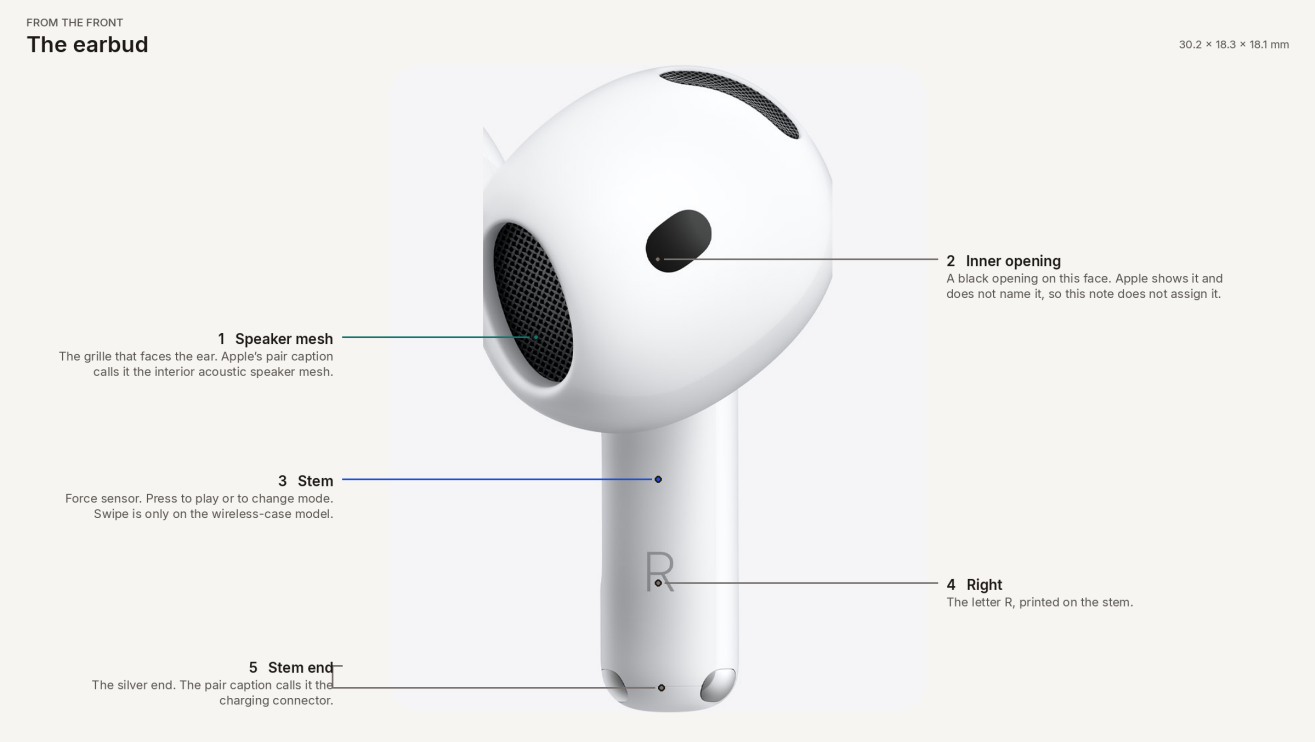
A research note on the regular AirPods, not AirPods Pro, written to the same depth as the Pro 3 dossier. Two versions share the earbud shape. The split is the case. Active Noise Cancellation is on both.

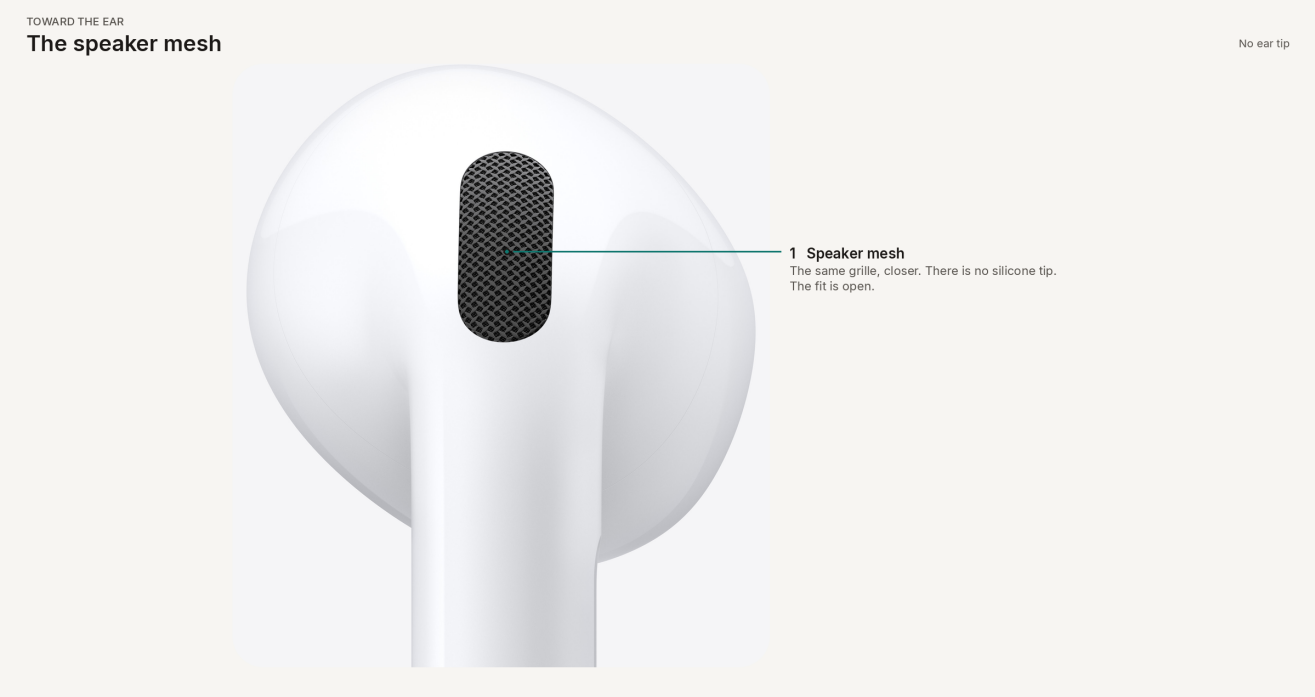
Announced 9 September 2026. In stores 18 September 2026. 129 US dollars with the USB-C case, 149 US dollars with the Wireless Charging Case. H2 in each earbud, Bluetooth 5.3, IP57 on the earbuds and the case.

Each earbud is 30.2 × 18.3 × 18.1 mm and weighs 4.3 g. Either case is 46.2 × 50.1 × 21.2 mm. The USB-C case weighs 32.3 g. The wireless charging case weighs 34.7 g. There is no ear tip. Firmware on both versions, from Apple's update page dated 17 September 2026, is 9A350. The USB-C earbuds are A3531, A3532, and A3533, with case A3530. The wireless-case earbuds are A3439, A3440, and A3441, with case A3529. Apple lists three earbud numbers per version and does not say which is left. The photographs print L and R on the stems. Nothing here was measured on hardware in this project. Where Apple does not publish a number, the line says so.

## What each part is

Each plate is one of Apple's own photographs or Apple's own interior drawing. A thin line runs from the label to the part. The black opening on the inner face is visible and unnamed, so it stays unnamed. The millimetre figures are Apple's published overall sizes, the same for both versions of the earbud.



THE STEM

## The control surface

No mechanical button




**2 Speaker mesh**
The grille on the side of the earbud.

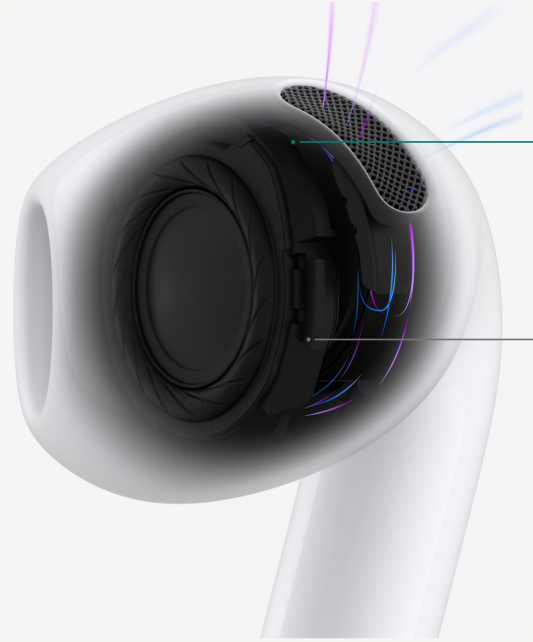
**1 Volume swipe**
The stem is the control. On the wireless-case model, a swipe along it changes the volume.

**3 Stem end**
The end of the stem. A close-up on Apple's page captions this opening as the external stem microphone.

APPLE'S INTERIOR DRAWING

## The driver

Open ear, no tip



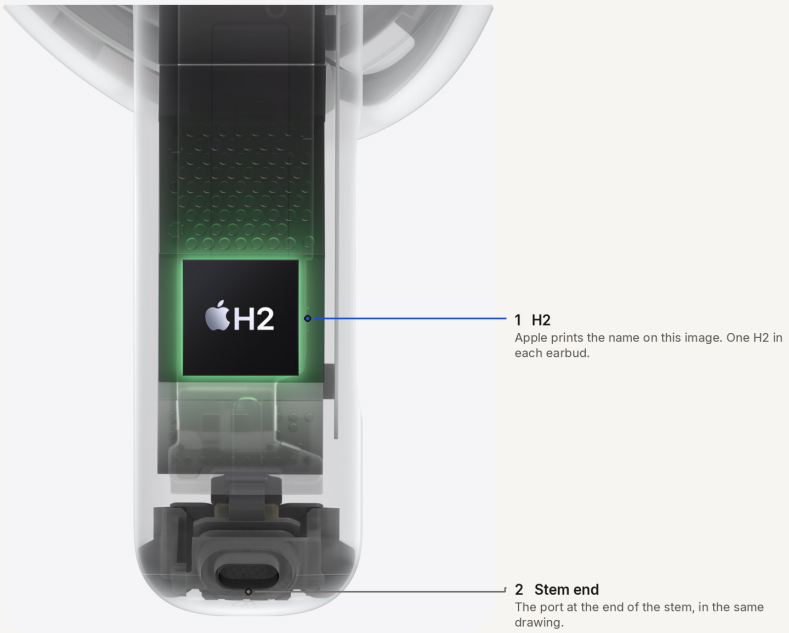
**2 Speaker mesh**
Apple's caption says the coloured lines are air and sound coming through the earbud. They are not a measurement.

**1 Driver**
The round speaker. Apple calls it a custom high-excursion driver. Diameter unpublished.

APPLE'S DRAWING

### Inside the stem

H2 in each earbud




CLOSED

### The case

46.2 × 50.1 × 21.2 mm




## How to read the figures

Blue is Active Noise Cancellation. Stone is the open fit with noise control off. This model has no foam seal, so stone is not a tip, and there is no teal row because Apple has not published a Transparency measurement. The laboratory curves are one fixture. They are a different kind of number from Apple's "up to 1.5×" sentence.

On the compare page, both AirPods 5 versions are "up to 1.5× more Active Noise Cancellation than AirPods 4 with Active Noise Cancellation," tested to IEC 60268-24. The newsroom says the same comparison as "up to 50 percent more." Those are one claim in two wordings. A separate sentence calls the result best in class for an open ear: Apple, July 2026, iPhone 17, prerelease firmware and iOS 27, IEC 60268-24, against the best-selling wireless open-ear headphones then on sale. On the same compare page, AirPods Pro 3 is "up to 3× more Active Noise Cancellation than AirPods 5," also IEC 60268-24. None of those sentences comes with a curve. A doubling of acoustic power is 3 dB. A sound that seems about half as loud is closer to 10 dB. This note does not rewrite 1.5×, 50 percent, or 3× as decibels.

### What active cancellation adds, by frequency

Stone is the open fit with ANC off. Blue is ANC on. There is no silicone tip, so the low end has almost no passive block.

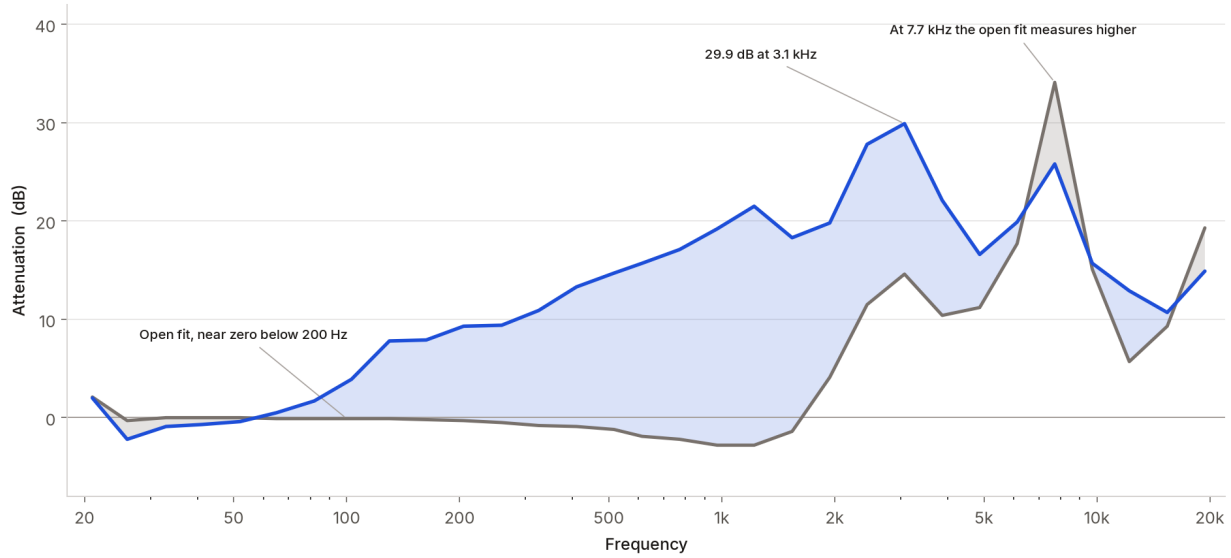



Figure 1 SoundGuys B&K 5128, AirPods 5 with Wireless Charging Case. Decibels of outside noise blocked. Not Apple's hearing-protection test. Fit changes an open earbud.

Below about 200 Hz the open fit blocks essentially nothing, and a few points sit slightly below zero. At 103 Hz, ANC on is 3.9 dB and ANC off is −0.1 dB, so the canceller is doing the low-frequency work the shape does not. The deepest ANC-on point on this sweep is 29.9 dB at 3.1 kHz. At 7.7 kHz the open fit alone measured 34.1 dB and ANC on measured 25.8 dB, so cancellation is not more blocking at every frequency. SoundGuys also reports a 59 percent reduction in perceived loudness, against 53 percent for AirPods 4 with ANC, plus scores of 2.9/10 isolation, 5.9/10 ANC, 3.7/5 MDAQS, and 6.4/10 editorial. Those are their metrics, not Apple's 1.5×. They note the buds change balance with volume, so one curve is not every volume. RTINGS has a review of this model. It is paywalled and was not opened, so this note has no RTINGS category table.

### Playback against a preference target

Zero is the SoundGuys target, not silence. The open fit drops the bass. The three-band Custom EQ is a separate control, and Apple does not publish its range.

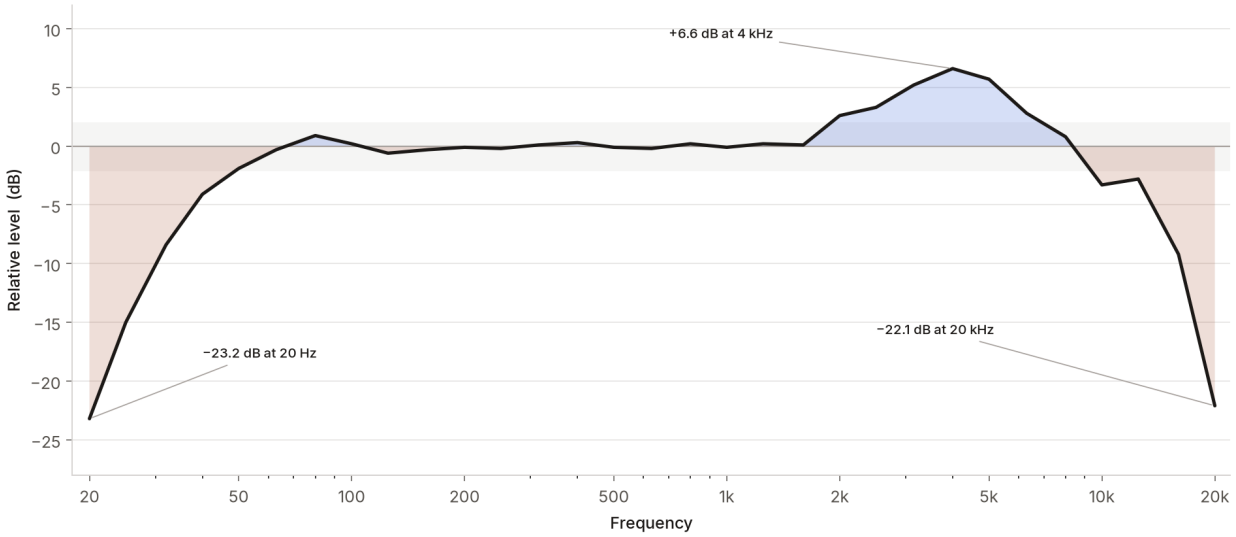



Figure 2 SoundGuys B&K 5128, relative to their preference curve. The same wireless-case model. Not an Apple specification.

From 80 Hz to about 1.6 kHz the curve sits within about 1 dB of the lab target. The bass does not: −8.4 dB at 32 Hz and −23.2 dB at 20 Hz. There is a lift of 6.6 dB at 4 kHz, then a fall to −9.2 dB at 16 kHz and −22.1 dB at 20 kHz. Adaptive EQ still runs from the inward microphone, and Apple does not publish its bands. Separately, a three-slider equalizer, Low, Mid, and High, is on AirPods 4 or later, so both versions have it. The path is Settings, the AirPods, Audio & Routing, Equalizer, Custom. It is stored on the earbuds. Recommended restores

Apple’s profile. Apple does not publish the decibel range, the center frequencies, or a Q. There is no 10-band table.

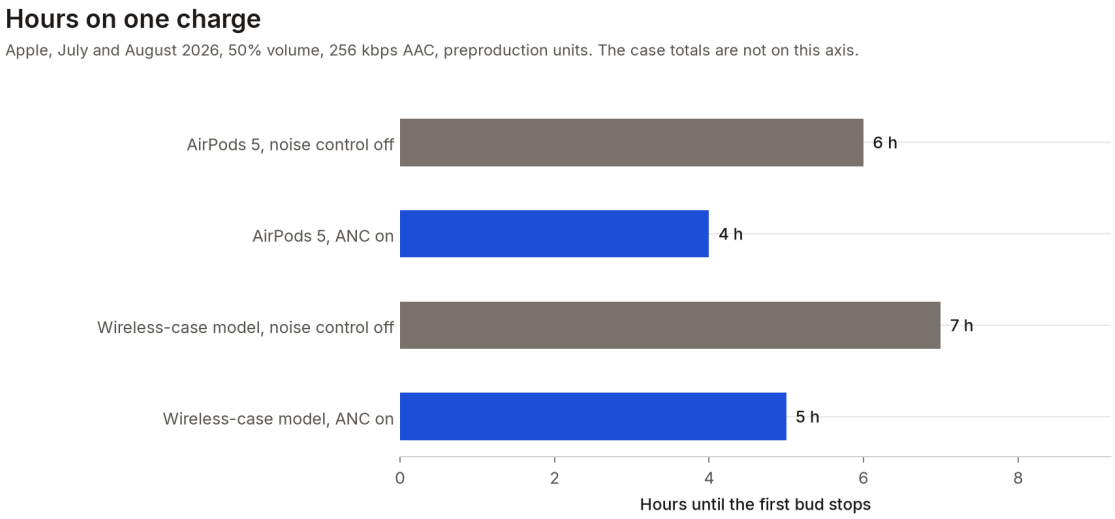



Figure 3 Stone is noise control off. Blue is ANC. Not drawn: 20 h with the USB-C case, 22 h with the wireless case, both with ANC. Five minutes in the case is about 1 h.

Apple’s test, July and August 2026, preproduction buds and a preproduction iPhone 18 Pro, 358 tracks at 256 kbps AAC, volume at 50 percent, Spatial Audio and Conversation Awareness off. The USB-C model is 4 hours with ANC and 6 hours with noise control off. The wireless-case model is 5 hours with ANC and 7 hours with noise control off, one hour longer on ANC than AirPods 4 with ANC. With the case those ANC totals are 20 hours and 22 hours, and they are not drawn on this axis. Five minutes in either case is about one hour with noise control off. Apple does not publish a noise-control-off total with the case, a talk-time figure, or a Transparency-only battery figure. The two cases are not interchangeable. Each fits only its own earbuds.

## Listening modes

Four states exist: Active Noise Cancellation, Transparency, Adaptive Audio, and Off. Press and hold cycles them. Either stem can be set to that cycle or to Siri, and the settings guide says which modes are in the cycle can be changed. The controls page names the hold as switching among ANC, Transparency, and Adaptive Audio. Off is in the listening-mode guide. This note does not add a rule that Off must be turned on first. That sentence is written for AirPods Pro 2 and AirPods Pro 3.

Apple’s noise-control article, which names both AirPods 5 models, describes the mechanism. Outward microphones hear noise heading toward the ear, and the earbud plays an opposing sound. An inward microphone hears what is left in the ear, and the earbud counters that too. That is feedforward plus feedback. There is no foam tip, so with the electronics off the only reduction left is the open shape, which Figure 1 shows is about zero under 200 Hz. Transparency is a processed feed of the outside world, not the same thing as Off. Adaptive Audio blends the two as the room changes. A slider under Listening Mode, Adaptive, or in Control Center, pushes the blend toward Transparency or toward Noise Cancellation. Apple does not publish that slider in decibels.

Conversation Awareness lowers media volume when the wearer starts speaking, and it comes back when they stop. Spoken audio, such as a podcast, pauses after a few seconds. It does not transcribe the conversation. Press and hold the stem to end it by hand. Personalized Volume learns a preferred loudness and adjusts for the room. Voice Isolation cleans the outbound voice on a call. None of the three has a published decibel figure. Loud Sound Reduction is not a control on this model. Apple’s guide limits it to AirPods Pro 2, AirPods Pro 3, and AirPods Max 2.

## What Pro 3 has, and this model does not

AirPods Pro 3 has an official per-frequency table: the unpowered seal, octave by octave, on Apple’s hearing-protection sheet, plus a passive NRR and a powered attenuation band. AirPods 5 does not. No ANSI datasheet, no NRR, no CSA class, and no octave-band mean turned up in the manuals list or the tech specs. Hearing Protection, the Hearing Test, the Hearing Aid feature, and Conversation Boost sit on the Pro columns of the compare page, not on these. Heart-rate sensing is a Pro 3 row. The optical in-ear sensor on AirPods 5 detects that the bud is in an ear. It is not a heart-rate sensor, and the black opening on the inner face stays unnamed because Apple’s caption does not name it. This note does not borrow Pro 3’s 26 to 32 dB hearing-protection band, its 12 dB NRR, or its octave table. Those belong to a sealed earbud.

## Sensors

Each earbud has two outward beamforming microphones and one inward microphone. Six channels across the pair. The outward pair is shared by the voice and by feedforward cancellation. The inward microphone is shared by feedback cancellation and by Adaptive EQ. The speech accelerometer detects that the wearer is talking. It is not a microphone. The motion accelerometer is head tracking, and the nod and the shake. The force sensor is the press. Volume swipe is on the wireless-case model only. The product page calls the grille the interior acoustic speaker mesh and the silver stem end the charging connector, and a separate close-up is captioned “external stem microphone.” Apple does not publish a drawing that separates those two jobs, so this note does not invent a second hole. There is no UWB chip in the case and no ear-tip fit sensor.

## Controls

Press once to play or pause, answer a call, mute or unmute, or use the camera remote. Press twice for the next track or to end a call. Press three times for the previous track. Mute and end-call can be swapped. Press and hold is the listening-mode cycle, or Siri, set per stem. Hold both stems, or say “Siri, start Live Translation”, or use the Action button, the Translate app, or Phone or FaceTime, for Live Translation. “Siri” or “Hey Siri” is the phone’s assistant. Nod or shake only if Announce Calls and Announce Notifications are on. Swipe for volume exists only on the wireless-case model. In Accessibility, that swipe can be turned off and its length can be changed. Apple does not publish how many decibels one swipe moves. Press Speed and Press and Hold Duration are on the same accessibility screen for AirPods 5. Apple does not publish milliseconds for either.

Take one earbud out, with Automatic Ear Detection on, and playback pauses. Put it back, or press the stem, and it resumes. Apple does not publish a delay in seconds. Take both out and playback stops and does not resume by itself. Turn detection off and audio keeps going to the buds whether they are worn. Noise Cancellation with One AirPod is supported: Settings, Accessibility, AirPods. Apple’s article names AirPods 4 with ANC, AirPods 5, and AirPods Pro. The microphone can be Automatically Switch, Always Left, or Always Right. Pause Media When Falling Asleep is a toggle for AirPods 5, under the AirPods name.

## Speech, and where it is decoded

The earbuds capture and play. The language models run on the phone. Siri and Siri AI are the phone’s assistant. Siri AI is a beta on iOS 27, macOS 27, and watchOS 27, English first, not initially in the EU on iOS, iPadOS, and watchOS, and not for users under 13. Hands-free access is described for all AirPods. Live Translation is on both AirPods 5 models, in beta, with Apple Intelligence, an iPhone 15 Pro or later, iOS 26 or later, and the latest firmware. Processing stays on the iPhone. Languages on the Apple Store buy page, read 5 November 2026: Chinese Mandarin Simplified and Traditional, English UK and US, French France, German Germany, Italian Italy, Japanese Japan, Korean South Korea, Portuguese Brazil, and Spanish Spain. Not every region. Generative output can be wrong. Conversation Awareness does not produce a transcript. Studio-quality recording is listed, and the product page ties the outward microphones to calls and to those recordings. Sample rate and bit depth are unpublished. The 9A350 note does not name Camera, Voice Memos, or Messages, so this note does not copy that app list from Pro 3.

## Charging and the radio

The USB-C case charges from USB-C. The wireless charging case charges from USB-C, an Apple Watch charger, or a Qi-certified charger. The charge guide lists MagSafe for other cases and does not list either AirPods 5 case. The wireless case has a speaker on the bottom for charging status and for Find My, and that speaker needs iOS 27 or later. The USB-C case's identify entry does not mention a speaker. Find My on both versions can show the buds and send a separation alert. Precision Finding is not described for these cases. The buds switch automatically among Apple devices on the same Apple Account, and a second pair can share audio by being brought near the iPhone, iPad, or Apple TV. Bluetooth 5.3. The battery test and the lab review both name AAC. The lab review says not aptX and not LDAC. Lossless over USB-C is not an AirPods 5 specification in this note. On a non-Apple phone they play as Bluetooth headphones, functionality may be limited, and the stem still presses. Writing ANC, Custom EQ, or the press-and-hold assignment from an Android app is not a supported Apple API.

## What this note does not claim

No decibel for 1.5×, for 50 percent, or for Pro 3's 3× against AirPods 5. No hearing-protection sheet, NRR, or octave table. No Hearing Aid, Hearing Test, Conversation Boost, Loud Sound Reduction, or heart rate. No driver diameter, impedance, sensitivity, published 20 Hz to 20 kHz window, sample rate, or bit depth. No range, center, or Q for the three equalizer sliders. No decibel per swipe and no millisecond for Press Speed. No resume delay, no pairing tap pattern, and no factory-reset gesture. No swapped cases, no ear tips, no UWB, no MagSafe. No RTINGS category averages. No Android API for ANC, EQ, or gestures. The photographs and plates are Apple's.